

PERCEPTUS & GESTUREUI

Voice Control, AI Screen Vision & Touchless OS Mastery

A comprehensive, accessible technical paper exploring natural voice control, intelligent AI screen understanding, and real-time machine vision powering the modern touchless desktop.

The Big Idea in Simple Words:

Perceptus frees you completely from the mechanical mouse and keyboard. You can simply look at your screen, point with your finger in mid-air, and speak naturally (such as "click this", "save the file", or "summarize what is on my screen"). An AI Screen Vision Brain comprehends what program is open, detects the active buttons, and executes the exact action on Windows for you!



Natural Voice Control

*Push-to-talk (**Ctrl+Space**) and continuous offline speech (Vosk) fused with spatial hand pointing for intuitive desktop commands.*



Intelligent AI Vision

Real-time screen comprehension extracts visual intent, locates UI targets, and synthesizes multi-step autonomous JSON Action Plans.

HOSTED SENSORY STUDIO & BRAIN

window-mind-whisper.lovable.app

WEB CONTROL COCKPIT

advancedlogiclabs.dpdns.org/gestures

COMPANION TACTILE PLATFORM (ATI)

advancedlogiclabs.dpdns.org/ati

ENGINEERING COLLECTIVE

Class of Learners • Neel, Ansh, Aarush & Yash Raj

01 Controlling With Voice: Speech-Driven Interaction

Voice control in Perceptus is not just a standard speech-to-text dictation box. It is an intelligent, low-latency operating system control loop that lets you command applications, manipulate windows, and type anywhere on Microsoft Windows using natural spoken speech.



1. Offline Voice Recognition (Vosk 16kHz)

Powered by `vosk-model-small-en-us-0.15` and `sounddevice`, speech is processed continuously on your local CPU without needing internet. It listens asynchronously with zero audio uploads, preserving total privacy.



2. Push-to-Talk (PerceptusFlow Ctrl+Space)

The desktop background pill utility (`PerceptusFlow.exe`) sits in the Windows taskbar. Holding **Ctrl+Space** activates instant listening mode, transitioning from *Ready* → *Listening* → *Thinking* → *Done*.

A. Spatial Voice + Gesture Fusion: The "Point and Say" Miracle

Standard voice assistants fail on desktops because they lack spatial context. If you say "click", a traditional assistant has no idea *where* to click. Perceptus solves this through **multimodal spatial fusion**:

- **Fusing Coordinates:** When you say "click", "double click", or "right click", the voice engine immediately polls the active gesture controller for your current hand pointer position `(cx, cy)`.
- **Natural Pointing:** You point your index finger at a video thumbnail, say "click", and the system executes a physical mouse click at that exact pixel location!
- **Voice Drag & Drop:** Point at a file, say "drag start", move your finger across the desk, and say "drop" to move the file effortlessly.

B. Direct Spoken Action Commands

Spoken Command	Speech Trigger Pattern	Executed Windows Action
"Click" / "Double Click"	Single word utterance	Clicks the mouse at your current pointing fingertip position.
"Scroll Up / Down"	Directional utterance	Smoothly scrolls the active document or web page vertically.
"Type [Message]"	Argument command: <code>"type hello"</code>	Simulates native keystrokes directly into the focused input box.
"Copy" / "Paste" / "Cut"	Clipboard voice shortcut	Executes native shortcuts: <code>Ctrl+C</code> , <code>Ctrl+V</code> , or <code>Ctrl+X</code> .
"Close Window" / "Maximize"	Window management	Executes <code>Alt+F4</code> to close or maximizes window via Win32.
"Open Keyboard"	Touch keyboard trigger	Pops up the touch keyboard (<code>TextInputHost</code>) or <code>keyboard.html</code> .



Spoken Text-to-Speech (TTS) Replies:

On the web cockpit at advancedlogiclabs.dpdns.org/gestures, Perceptus doesn't just listen—it talks back! Using browser speech synthesis, the brain vocally answers your queries, reads summaries out loud, and confirms executed commands.

Simple voice recognition can only execute pre-programmed keywords. The real magic of Perceptus comes from its **AI Cognition Brain**. The AI acts like an intelligent co-pilot sitting beside you, looking at your screen, listening to your goal, and figuring out what needs to be done.

A. Screen Understanding via AI Vision (Sensor 02)

Whenever you give a high-level command like "save my work" or "export this video", the AI takes a downscaled desktop screenshot (1280px JPEG) and inspects the visual interface:



1. Recognizing Open Programs

The AI identifies what software is open (e.g. VS Code, Premiere Pro, Chrome, or Blender) and reads window titles and tab headers to know where you are.



2. Detecting Visual Buttons & Text

The AI locates buttons, input boxes, timeline playheads, and menus with exact coordinate boxes, so it knows precisely where to target mouse clicks.

B. Turning Vague Human Speech into Exact Action Plans (JSON DSL)

Human beings speak in goals, not technical steps. You might say: "make this louder" or "save my work and exit". The AI Cognition Brain translates that abstract goal into an executable **Action Plan**:

COGNITION REASONING: HOW THE AI DECIDES WHAT TO DO

JSON DSL

```
// USER SAYS: "Save this document and close it"
// AI EXAMINES SCREEN: Notepad is open with title "Meeting_Notes.txt *" (unsaved)
{
  "understanding": {
    "app": "Notepad",
    "situation": "Document has unsaved changes marked with asterisk (*)"
  },
  "intent": { "label": "save_and_exit", "confidence": 0.96 },
  "plan": {
    "rationale": "Press Ctrl+S to commit notes to disk, then Alt+F4 to exit cleanly.",
    "actions": [
      { "op": "hotkey", "target": "Notepad", "keys": ["ctrl", "s"] },
      { "op": "wait", "amount": 250 },
      { "op": "hotkey", "target": "Notepad", "keys": ["alt", "f4"] }
    ]
  },
  "say": "Saved your meeting notes and closed Notepad."
}
```

C. Action Plan Operations (What the AI Can Drive)

- **move** / **click**: Moves the hardware mouse cursor to exact normalized coordinates **[x, y]** and clicks.
- **hotkey** / **key**: Simulates keyboard shortcuts (like **Ctrl+S**, **Alt+Tab**, **Ctrl+K**, or **Enter**).
- **type**: Types out full sentences, URLs, or code blocks into text editors and search bars.
- **scroll**: Automatically scrolls up or down through long PDFs or web pages.
- **launch** / **focus_window**: Opens installed applications or brings background windows to the foreground.

03 Architecture & The Windows Desktop Agent Fleet

Perceptus connects the web and your desktop operating system through a seamless 4-tier pipeline:

The 4-Tier Pipeline:

1. **Perception:** In-browser MediaPipe WASM/GPU tracks 468 face points & 21 hand joints.
2. **Cognition Brain:** Vision AI at `/api/public/brain` synthesizes intent & creates Action Plans.
3. **Memory Vault:** Remembers your name, favorite apps, and multi-turn conversation history.
4. **OS Execution:** Local Python agents execute physical clicks, smooth cursor motion, and key presses on Windows.

Meet the Windows Desktop Agent Suite

Desktop Agent	Technologies	What It Does On Your Computer
<code>PerceptusFlow.exe</code>	PyStray, Global Hotkeys	A floating status pill in the system tray. Hold <code>Ctrl+Space</code> to talk, and it carries out actions or dictates text into any active app.
<code>perceptus_desk.py</code>	Tkinter, OpenCV, PIL	Desktop window with live webcam preview. Fixes browser camera bugs by testing <code>DirectShow</code> → <code>MediaFoundation</code> → <code>Native</code> across cameras 0-4.
<code>perceptus_pilot.py</code>	PyAutoGUI, ScreenGrab	Autonomous helper. Takes a screenshot, queries the brain, executes steps, and continues multi-step goals until completed.
<code>mouse_server.py</code>	WebSockets (Port 9999), PyWin32	Real-time gesture controller. Warps cursor via <code>user32.SetCursorPos</code> with 2D Kalman filtering for zero lag.
<code>main_app.py</code>	PyWebView 6.2, Vosk	All-in-one desktop application bundling the web interface, WebSocket server, and offline voice engine into one executable.

Kinematic Smoothing: Kalman Filter & Touch Keyboard

- **2D Kalman Filter:** Eliminates camera hand jitter. It dynamically adjusts its noise equations (`Q = 0.05` for steady pointing, `Q = 0.12` for fast sweeps) and `R = 0.8` so pointing feels smooth like an iPad stylus.
- **Smart Touch Keyboard:** When you want to type without touching the desk, Perceptus launches Windows 11 `TextInputHost.exe`, falls back to Windows 10 `TabTip.exe`, or opens the custom cyber-styled `keyboard.html`.

04

Memory, Safety Guarantees & The Live Web Portals

Having an AI that can move your mouse and type on your computer requires safety, long-term memory, and simple web access.

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1. Persistent Memory Vault

Perceptus remembers across sessions. When signed in with Supabase, it extracts facts (your name, preferred applications, default folder paths) and maintains multi-turn conversation history. Anonymous mode runs safely in temporary RAM.
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2. Failsafes & Confirmation Gates

Corner Slam: Slamming the mouse to any corner immediately kills automation via `pyautogui.FAILSAFE`.

Confirmation: Risky steps (deleting files, closing unsaved apps) are paused until you click confirm.

The Online Portals Powering Perceptus

Online Portal URL	Platform Role	Core Features & User Experience
window-mind-whisper.lovable.app	Cloud Perception & Cognition Engine	Live webcam sensor studio running face mesh & hand tracking, screen understanding playground, Brain API documentation, and Windows app downloads.
advancedlogiclabs.dpdns.org/gestures	Class of Learners Control Hub (<code>gesture.html</code>)	High-contrast Cyber-Dark telemetry cockpit, voice recognition microphone button, live latency monitoring, TTS audio replies, and download hub.
advancedlogiclabs.dpdns.org/ati	Companion Tactile Flagship (ATI v2.5.0)	The tactile mechanical sister-project: 200-level deterministic typing curriculum, real-time multiplayer racing rooms, and PKCE Google sign-in.

Synergy with Advanced Typing Instructor (ATI):
Class of Learners explores both extremes of human-computer interaction: **ATI** trains muscle memory for blazing-fast 120+ WPM typing on physical keyboards, while **Perceptus** provides the ambient touchless future where voice and machine vision do the heavy lifting!

Real-World Scenario: Voice & AI Cognition in Action

- ### 1. You Speak & Point

You point at a document and say: "Summarize this and send it in Slack." Microphone streams audio while MediaPipe captures your fingertip vector.
- ### 2. AI Screen Inspection

Sensor 02 captures a 1280px frame of your screen, reads text via OCR & Vision LLM, and generates a concise summary in working memory.
- ### 3. Action Plan Execution

The brain compiles an Action Plan: focuses Slack via Win32, types the summary into the channel, and simulates `Enter`.
- ### 4. Vocal Confirmation

Perceptus plays a cyber chime and speaks back through your speakers: "Summary posted to Slack!" Total interaction time: under 2 seconds.

Complete Technology Inventory & Final Vision

Here is the complete inventory of all technologies, tools, and protocols used across the Perceptus and Class of Learners ecosystem:

Domain	Tools & Libraries Used	Role in the System
Vision & AI	MediaPipe Hands & FaceMesh, OpenCV, Vision LLM API, Tesseract.js	In-browser joint landmark tracking, multi-backend camera capture, cloud screen understanding, OCR air-drawing.
Desktop OS Control	Python 3.10+, PyWebView 6.2, PyAutoGUI, PyWin32 (user32), PyGetWindow, PyStray	Direct Win32 cursor positioning (<code>SetCursorPos</code>), window moving, hardware mouse clicks, system tray pill.
Voice & Audio	Vosk (small-en-us-0.15), SoundDevice, Web Speech API, Web Audio API	Continuous offline speech recognition, push-to-talk audio streaming, browser speech transcription, TTS replies.
Networking & Data	AsyncIO WebSockets, Flask, SQLite (<code>typing_quest.db</code>), Supabase, Firebase	Real-time bi-directional coordinates (port 9999), webhook bridge (port 8765), persistent facts and multiplayer sync.
Frontend & Design	HTML5 Canvas, CSS3 Cyber Tokens, Caveat & JetBrains Mono, Vanilla JS	High-contrast cyber-dark interface, live telemetry indicators, particle animations, and fluid responsive layouts.

Summary & Forward Horizon:

Perceptus proves that personal computing doesn't have to stay stuck in the past. By fusing **voice control**, **hand pointing**, and **AI screen understanding**, any desktop user can enjoy effortless, touchless productivity. Future upgrades will bring native WebGPU 120 FPS tracking, on-device small vision language models (SLMs) for completely offline operation, and universal macOS/Linux support.

Class of Learners (CoL) • Advanced Logic Labs

Engineered by Neel, Ansh, Aarush & Yash Raj • Redefining human perception and desktop intelligence.

Explore: window-mind-whisper.lovable.app • advancedlogiclabs.dpdns.org/gestures • advancedlogiclabs.dpdns.org/ati